Machine Learning Model

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Bootcamp BI Batch 18

BUSINESS QUESTION ?

- What Factors Cause Customer Churn as a Preliminary Assumption?
- Early Assumption:

  1. Customers with monthly contracts are at higher risk of churn.

  2. Customers with high costs are more likely to churn.

  3. New customers churn more frequently.

- Does Churn Increase or Decrease?
- Can a Churn Prediction Model Be Created and Interpreted? How Well Does the Model Predict?

WHY SUPERVISED LEARNING AND WHY CHOOSE LOGISTIC REGRESSION AND RANDOM FOREST CLASSIFICATION ?

I use Supervised Learning because the dataset already contains clear labeled targets (Churn / Not Churn), which allows me to directly build a predictive model without first exploring latent patterns as in Unsupervised Learning. Since the problem is a binary classification task and requires interpretable probabilistic outputs, Logistic Regression is chosen as a baseline model due to its simplicity and ease of interpretation for non-technical stakeholders.

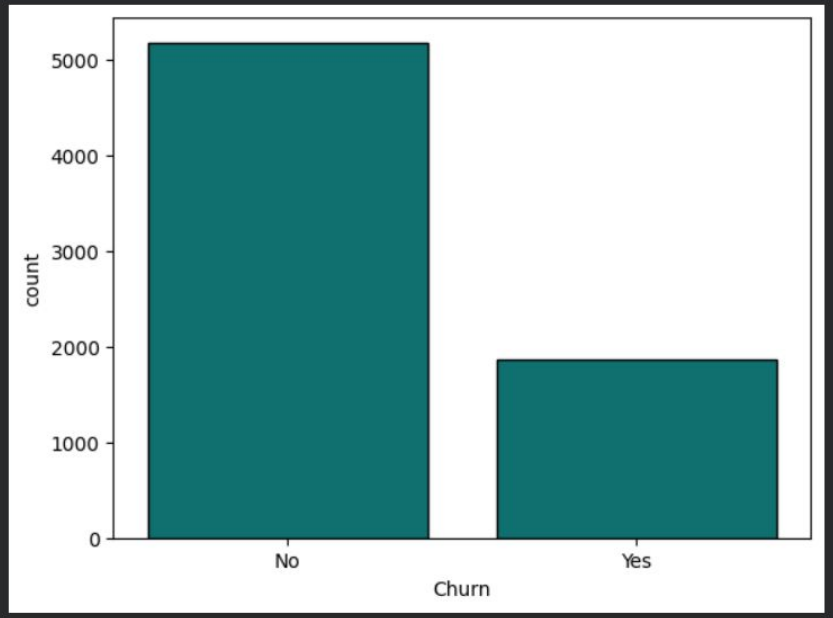
However, because Logistic Regression assumes a linear relationship in the log-odds, I also use Random Forest Classification as a comparison model. Random Forest is more capable of capturing non-linear patterns and complex feature interactions, allowing me to evaluate the trade-off between model interpretability, predictive performance, and business strategy considerations.

Data Cleaning

In the data cleaning stage, I used Python to explore the basic characteristics of the dataset, including identifying missing values and checking data types. I found that the `TotalCharges` column was incorrectly stored as an object instead of a numerical type, so I converted it to float. Some missing values appeared due to customers with zero tenure, where `TotalCharges` is logically equal to zero. Therefore, I filled these missing values with 0 to maintain data consistency and to avoid issues during EDA, modeling, and mathematical computations.

Exploratory Data Analysis (EDA)

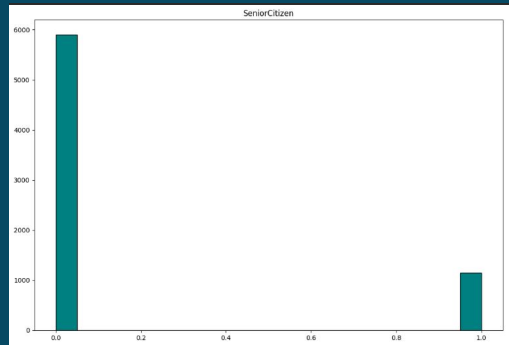
```
sns.countplot(x='Churn', data=df, color='Teal', edgecolor='black')
plt.show()
```

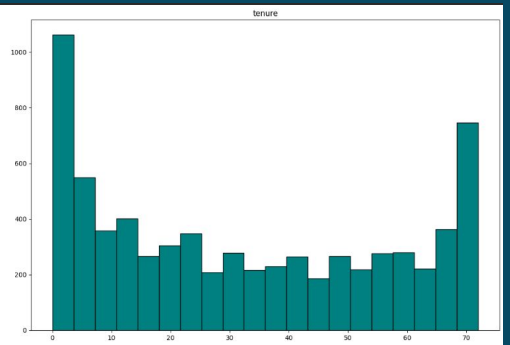


During the EDA process, I found that the number of churn customers was significantly smaller than non-churn customers, indicating an imbalanced dataset. This is important because the model will learn from this imbalance, and the prediction threshold can later be adjusted to support different business strategies, such as prioritizing recall to identify high-risk churn customers.

Exploratory Data Analysis (EDA)

Although the main objective of this project is to build a prediction model, I still conducted Exploratory Data Analysis (EDA) to understand the distribution of non-target variables.

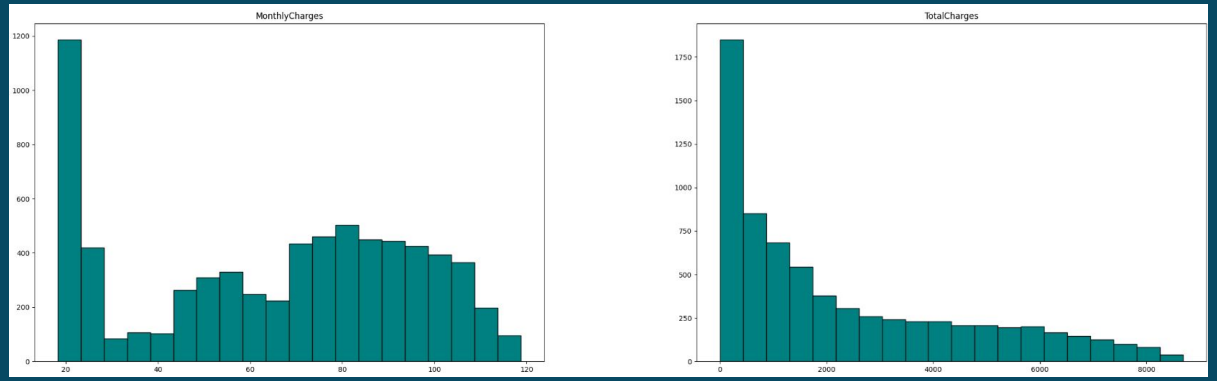




From the visualization, most customers are non-senior citizens, indicating that the customer base is dominated by a younger age group. In terms of tenure distribution, there is a high concentration of customers in the lowest tenure group, followed by a fluctuating pattern in the middle tenure range, and another relatively high concentration in the highest tenure group.

This pattern suggests two dominant segments: new customers who have just joined and long-tenured customers who are likely more loyal. Customers in the early tenure stage may indicate a higher potential risk of churn, while customers with high tenure reflect a stable and trusting customer base that continues to use the telco services.

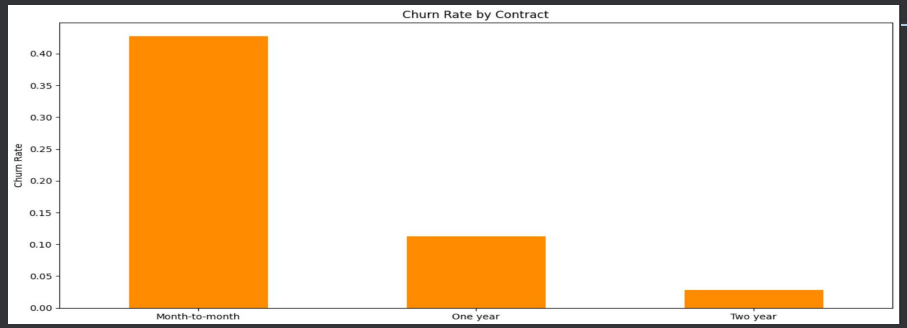
Exploratory Data Analysis (EDA)

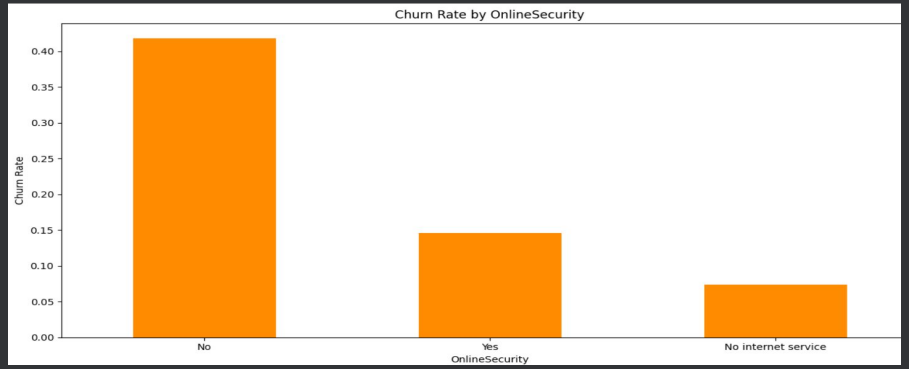


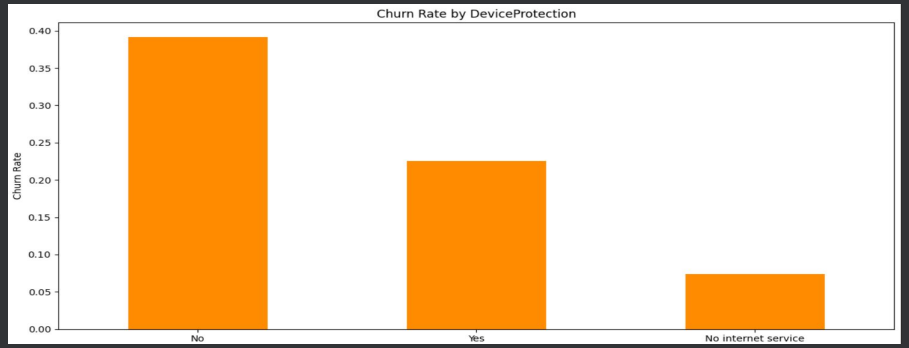
In this EDA, I focused on Monthly Charges and Total Charges, as both variables are closely related to tenure. The distribution of Monthly Charges shows a spike at lower values, followed by a more fluctuating pattern in the middle range and a decreasing trend at higher charges.

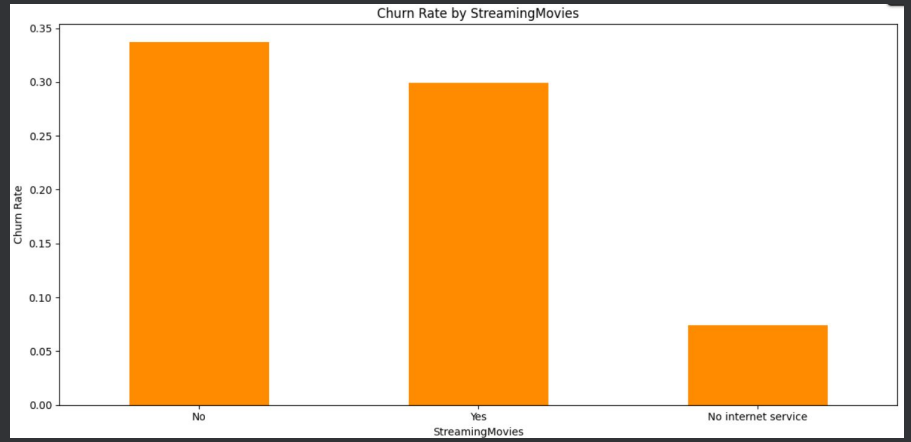
This may indicate that customers tend to maintain relatively lower spending levels over time or discontinue service before reaching higher total charges. Therefore, despite the presence of loyal customers, the risk of churn remains, especially when customer spending does not show sustained growth.

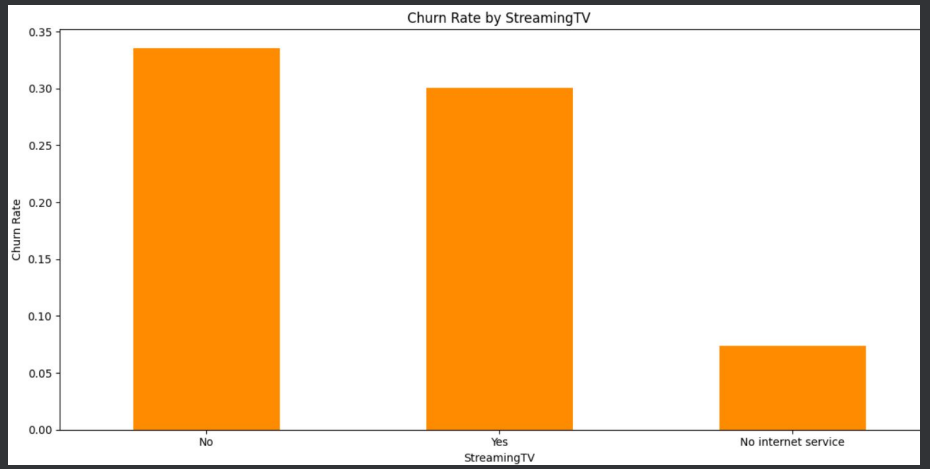
Meanwhile, Total Charges exhibit a declining distribution as the value increases, with the highest concentration appearing in the lowest cost group. When this pattern is linked to the previous tenure analysis, it suggests that while the telco has a segment of loyal customers, most customers do not accumulate very high total spending.


Churn Rate by Contract
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Month-to-month
One year
Two year


Churn Rate by OnlineSecurity
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
No
Yes
No internet service
OnlineSecurity


Churn Rate by DeviceProtection
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
No
Yes
No internet service


Churn Rate by StreamingMovies
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
No
Yes
No internet service
StreamingMovies


Churn Rate by StreamingTV
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
No
Yes
No internet service
StreamingTV

EDA Interpretation – Categorical Features and Churn Rate

Overall, this EDA indicates that customer churn is influenced by multiple factors rather than a single feature, reinforcing the need for a predictive model that can capture interactions between contract type, service usage, and customer behavior.

Another important observation is the presence of churn among customers with **No Internet Service**. This category appears across multiple features and may introduce noise if treated the same as regular service users. Therefore, during the modeling stage, this category may need to be handled separately or carefully encoded to allow the model to learn more precise churn patterns.

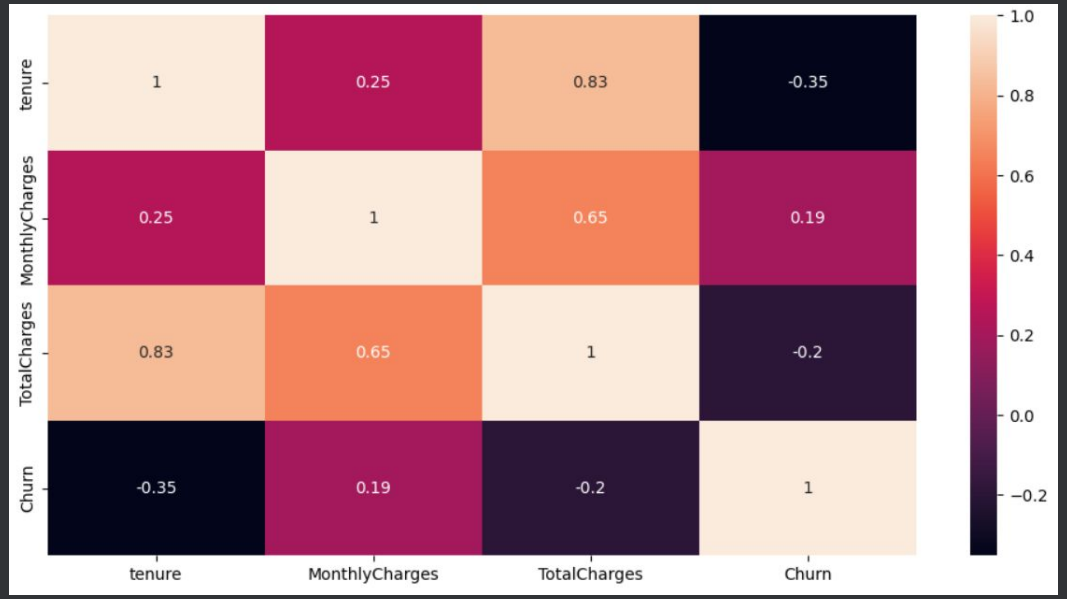
Interestingly, for **Streaming Movies** and **Streaming TV**, the churn rates between customers who use these services and those who do not are relatively similar. This may indicate that these entertainment features do not provide strong added value in terms of reducing churn, or that their perceived benefit is not sufficient to significantly influence customer loyalty.

This is the final EDA stage before conducting correlation analysis. In this step, I explored several non-target categorical features by applying simple mathematical expressions to estimate churn rate probabilities. Although not all features were included, this analysis provides an initial overview of how different products, services, and contract types may influence customer churn.

From the visualization, customers with **month-to-month contracts** show a noticeably higher churn rate compared to those with **one-year or two-year contracts**. This supports the earlier EDA insight related to tenure, indicating that early-stage customers tend to have a higher risk of churn.

In other aspects, such as **Device Protection** and **Online Security**, customers who do not use these services exhibit higher churn rates than those who do. This suggests that additional protection services may contribute to customer retention.

Correlation



Tenure shows a negative correlation with churn (-0.35), indicating that customers who stay longer tend to have a lower likelihood of churning.

Monthly Charges exhibit a weak positive correlation with churn (0.19). While the correlation is not strong enough to suggest a dominant influence, it indicates that higher monthly costs may slightly increase the risk of churn.

Total Charges show a weak negative correlation with churn (-0.20). This suggests that customers who have accumulated higher total spending tend to churn less, although this relationship is weaker than tenure. This may be because total charges are strongly influenced by tenure itself, making tenure a more direct indicator of customer retention behavior.

Overall, the correlation results indicate that no single numerical feature strongly determines churn on its own. Instead, churn behavior is likely influenced by a combination of customer tenure, pricing, and service-related factors, which justifies the use of multivariate machine learning models rather than relying solely on correlation analysis.

	precision	recall	f1-score	support
0	0.85	0.89	0.87	1035
1	0.66	0.57	0.61	374
accuracy			0.81	1409
macro avg	0.75	0.73	0.74	1409
weighted avg	0.80	0.81	0.80	1409

On the other hand, if the business prioritizes identifying as many churn-prone customers as possible, recall becomes more critical, even if it leads to higher false positive rates.

The F1-score provides a balanced evaluation by combining precision and recall into a single metric, making it useful when both types of errors are considered equally important. From a mathematical perspective, it helps assess how well the model balances these two competing objectives.

Machine Learning Model For Logistic Regression

The interpretation of this classification report depends heavily on the business perspective and stakeholder priorities. If the company aims to minimize the negative impact of false positives—such as incorrectly flagging loyal customers as churn—then precision becomes an important metric, although this may come at the cost of missing some actual churn cases (false negatives).

Machine Learning Model For Logistic Regression

A noticeable gap between macro average and weighted average metrics suggests that the model performs significantly better on the majority class (non-churn) than on the minority class (churn). This indicates the presence of class imbalance and a potential prediction bias toward non-churn customers.

Additionally, since the model uses a default probability threshold of 0.5, its predictions may not be optimal for detecting churn cases. Therefore, further evaluation using ROC–AUC analysis and threshold adjustment is necessary to align the model’s performance with specific business strategies.

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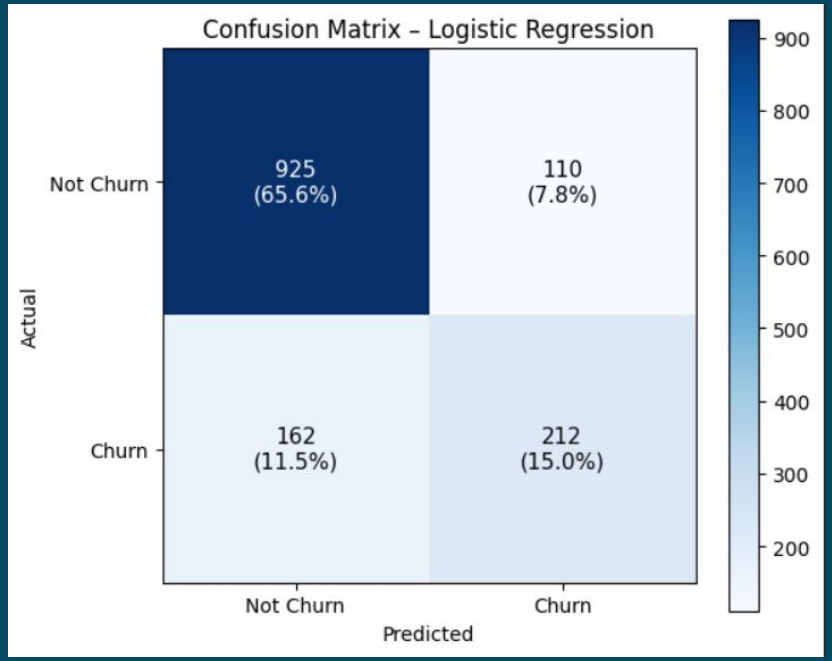


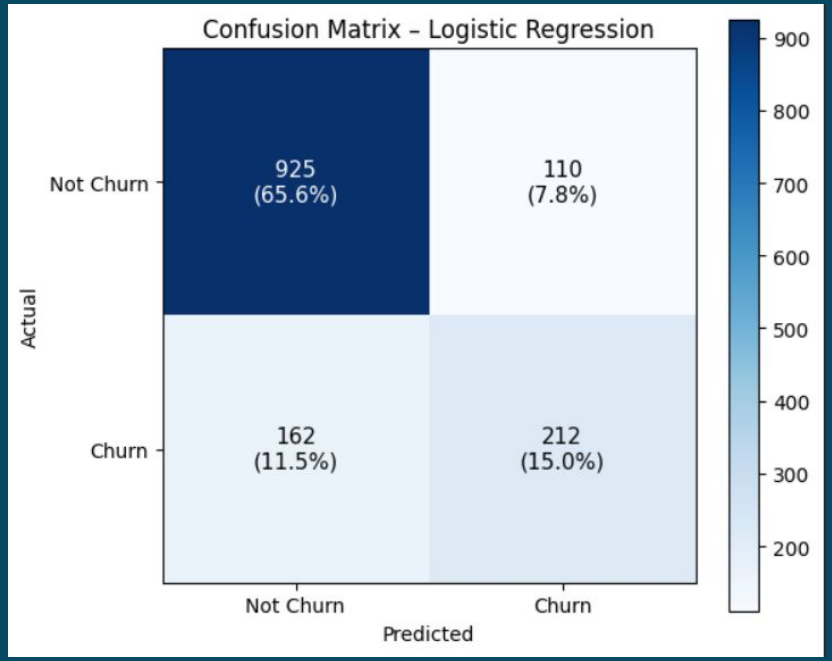
Based on the confusion matrix, the model correctly predicts a large portion of non-churn customers, indicating strong performance on the majority class. However, the number of correctly predicted churn customers is relatively lower compared to non-churn predictions.

A key aspect to consider is the trade-off between false positives and false negatives. The model produces fewer false positives than false negatives, suggesting that it is conservative in predicting churn and tends to avoid incorrectly labeling loyal customers as churners.

While this behavior can be beneficial when the cost of false positives is high, it also implies that a significant number of actual churn customers are not being detected. From a business perspective, this represents a potential loss in retention opportunities.

Therefore, the current model performance should be evaluated based on stakeholder priorities. If the business prefers to minimize unnecessary retention costs, this model may be acceptable. However, if the goal is to capture as many churn-prone customers as possible, further threshold tuning or alternative models should be considered.

Machine Learning Model For Logistic Regression



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Random Forest Classification

In this Random Forest model, I applied a balanced class weighting instead of the default setting, which allows the model to pay more attention to the minority class (churn) rather than being dominated by the non-churn class. As a result, the model becomes more sensitive in detecting churn customers.

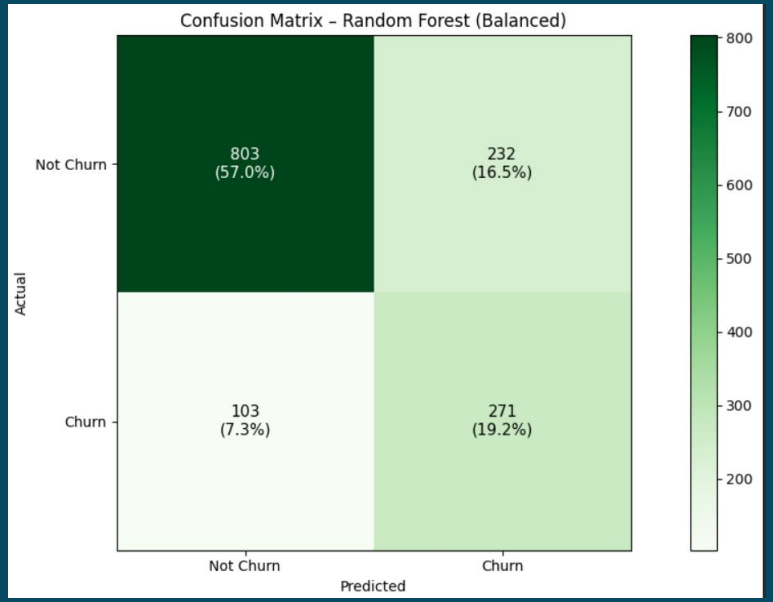
Compared to Logistic Regression, this model shows higher recall for churn, indicating that it is more effective in capturing customers who are likely to churn. However, this improvement comes with a noticeable decrease in precision, meaning that the model produces more false positive churn predictions.

	precision	recall	f1-score	support
0	0.89	0.78	0.83	1035
1	0.54	0.72	0.62	374
accuracy			0.76	1409
macro avg	0.71	0.75	0.72	1409
weighted avg	0.79	0.76	0.77	1409

This behavior reflects a clear trade-off between precision and recall. The Random Forest model tends to prioritize reducing false negatives, which is beneficial when the business impact of missing churn customers is high. However, it may also increase unnecessary retention efforts toward customers who would not churn.

Therefore, the choice between this model and Logistic Regression should depend on management priorities, whether the business prefers to minimize churn risk or control retention costs.

RANDOM FOREST CONFUSION METRIX

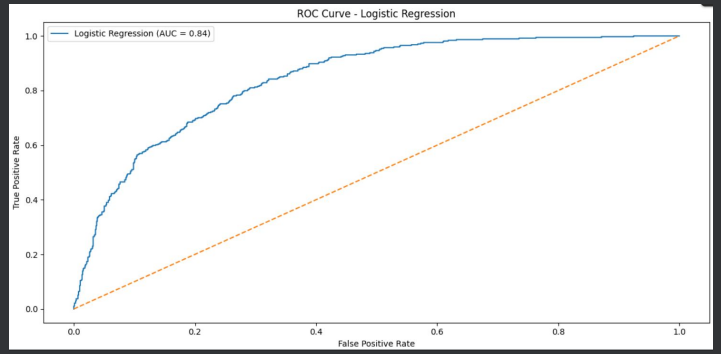


From the confusion matrix visualization, this Random Forest model further validates the previous evaluation results. The model clearly prioritizes reducing false negatives, meaning it is more aggressive in detecting churn customers.

This behavior is driven by the use of balanced class weighting and an ensemble of decision trees with controlled depth and a large number of estimators, which allows the model to better capture patterns from the minority class.

As a consequence, while the number of missed churn customers decreases, the model produces a significantly higher number of false positives compared to Logistic Regression. This indicates a trade-off where the company may incur higher retention costs by targeting customers who may not actually churn. Therefore, this model is more suitable when the business strategy prioritizes minimizing churn risk over controlling false alarm costs.

ROC - AUC LOGISTIC REGRESSION

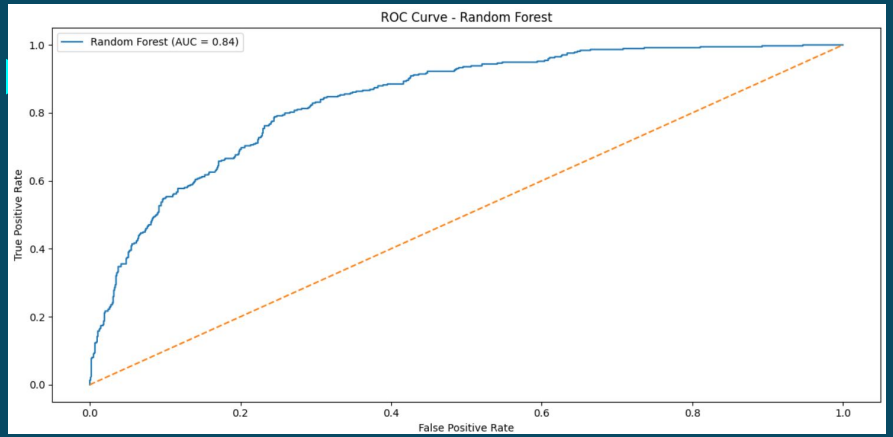


Based on the ROC curve visualization, the model demonstrates a strong ability to distinguish between churn and non-churn customers, as reflected by an AUC value of 0.84. This indicates that the model performs significantly better than random guessing.

From my interpretation, the curve shows that at certain threshold levels, a relatively small increase in the false positive rate can lead to a substantial increase in the true positive rate. In practical terms, this means the model is capable of identifying a large proportion of actual churn customers while only misclassifying a limited number of non-churn customers, especially in the early part of the curve.

However, management should be cautious when adjusting the decision threshold. If the threshold is lowered too aggressively, both the true positive rate and false positive rate will increase, which may result in excessive false alarms and inefficient campaign targeting. On the other hand, setting the threshold too conservatively may cause both rates to drop toward zero, leading to missed churn cases and an ineffective retention strategy.

Therefore, the optimal threshold selection depends heavily on business objectives and campaign strategy. This reflects the inherent trade-off between recall and precision, where improving the ability to capture more churn customers often comes at the cost of misclassifying more non-churn customers. The final decision should balance model performance with operational and business considerations.



ROC – AUC RANDOM FOREST CLASSIFATION

From a visual perspective, the ROC curve of the Random Forest model appears more convex compared to the Logistic Regression model, with a steeper rise near the lower false positive rate region. This suggests that the Random Forest model is able to achieve relatively higher true positive rates while maintaining low false positive rates, especially at certain threshold ranges.

The curve also appears more fluctuative, which is expected given the ensemble nature of Random Forest models, where predictions are aggregated from multiple decision trees rather than following a smooth linear probability function. This behavior indicates a more flexible decision boundary compared to Logistic Regression.

From a business perspective, this characteristic can be advantageous, as it allows management to identify a significant portion of potential churn customers without excessively increasing false alarms. However, this advantage is still highly dependent on management strategy and threshold selection.

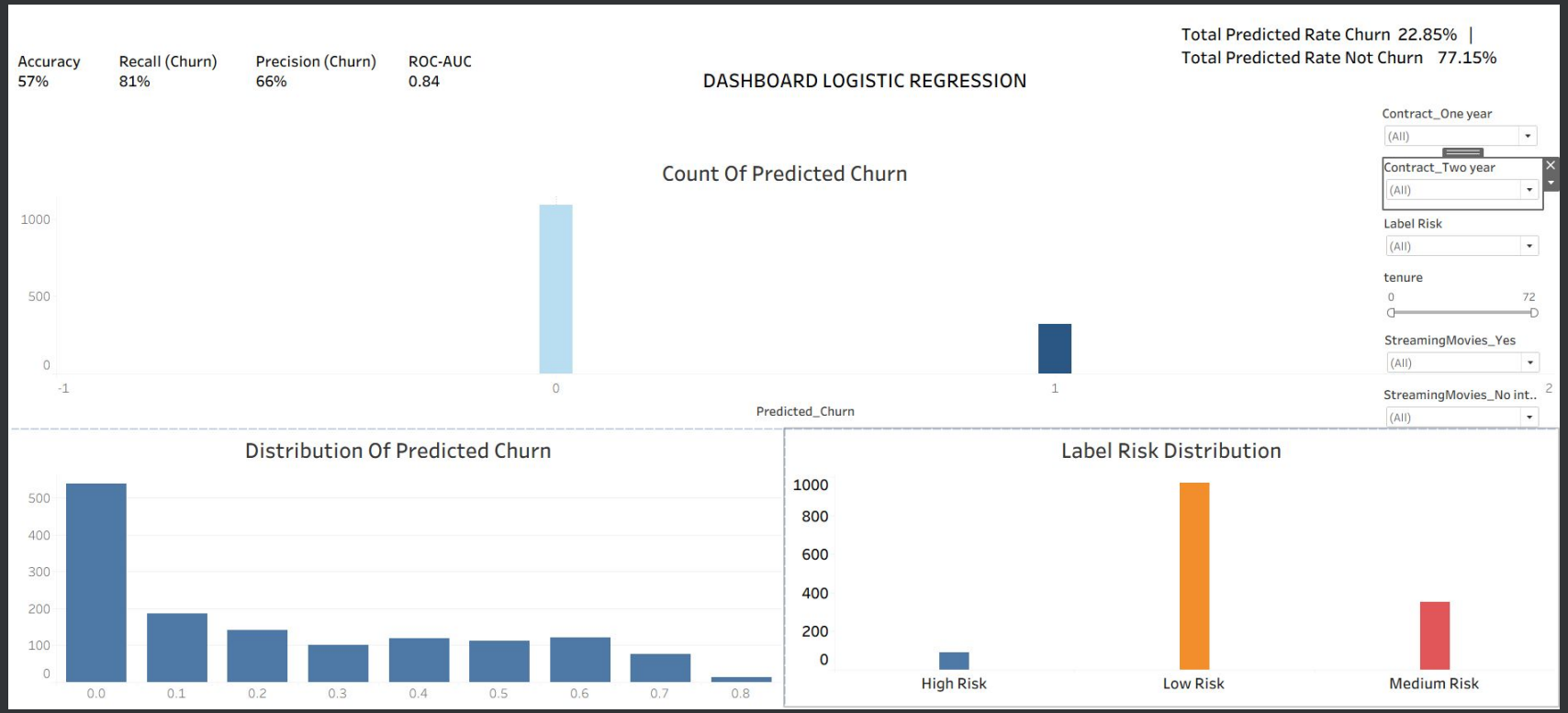
Similar to the previous model, management cannot afford to be overly aggressive or overly conservative when adjusting the decision threshold. An aggressive approach may lead to excessive false positives and increased operational costs, while a conservative approach may result in minimal churn detection and missed retention opportunities. Therefore, despite the stronger visual convexity, the Random Forest model still requires a careful trade-off between recall and precision aligned with business objectives.

WHY CHOOSE LOGISTIC REGRESSION OVER RANDOM FOREST CLASSIFICATION ?

Although both Logistic Regression and Random Forest achieved the same ROC–AUC score of 0.84, Logistic Regression was selected for further analysis and Tableau visualization. From a business intelligence perspective, Logistic Regression provides higher interpretability and more stable probability outputs, making the results easier to explain and use in managerial decision-making.

Compared to Random Forest, its predictions are less fluctuative and more suitable for consistent segmentation and threshold-based analysis in dashboards. Additionally, Logistic Regression carries a lower risk of overfitting when model tuning is limited, which supports more reliable performance on unseen data. Therefore, Logistic Regression was chosen as the final model, while Random Forest was used only as a performance benchmark.

VISUALIZATION WITH TABLEAU



Interpretation, Business Insight, and Recommendation

The visualization shows that the Logistic Regression model predicts a significantly larger number of customers as **non-churn (0)** compared to those predicted as **churn (1)**. This indicates that the majority of customers in the dataset are considered stable, while a smaller but meaningful segment is identified as having a higher risk of churn. From a business perspective, this imbalance is expected in churn problems and suggests that the model is not overestimating churn but rather focusing on customers with clearer risk signals.

The key insight from this result is that the predicted churn group represents a **targetable high-risk segment**. Even though its size is smaller, this group should be considered strategically important because customers predicted as churn are more likely to leave if no intervention is applied. Treating all customers equally would be inefficient; instead, this segmentation enables the business to prioritize retention efforts where they are most needed.

Based on this insight, the recommended business action is to design **focused retention campaigns** for the predicted churn group, such as personalized offers, loyalty incentives, or proactive customer outreach. Meanwhile, customers predicted as non-churn should be maintained with standard engagement strategies to avoid unnecessary retention costs. This approach supports cost-efficient decision-making by aligning intervention intensity with churn risk as identified by the model.

Reflective Question

Choosing the appropriate Machine Learning method and evaluation metrics before modeling is crucial because each algorithm and metric reflects different assumptions and business objectives. Testing many models without understanding the context often leads to superficial trial-and-error and results that look good numerically but lack business relevance. In churn prediction, metrics such as recall, precision, and ROC-AUC are more meaningful than accuracy alone, as the main goal is to identify at-risk customers. Early and thoughtful model selection leads to clearer interpretation and more reliable decision-making.

Data cleaning, preparation, and exploratory data analysis strongly influence model performance because Machine Learning models learn patterns directly from the data. Poor-quality, biased, or inconsistent data can produce misleading models regardless of algorithm complexity. EDA helps reveal distributions, class imbalance, relationships, and noise. Skipping or rushing this stage risks overfitting, unstable predictions, and incorrect business conclusions.

Data visualization and clear interpretation of model evaluation help non-technical stakeholders understand whether a model is suitable for use. Visual tools such as ROC curves, confusion matrices, and churn distributions communicate performance and trade-offs without requiring technical knowledge. In a business context, this enables management to assess risk, choose appropriate strategies, and confidently use the model to support data-driven decisions.

Link Tableau :

15′

https://public.tableau.com/app/profile/magasa.faiz/viz/Logistic_Regression_TelcoCustomerChurn/DashboardLogisticRegression?publish=yes

Link Collab :

https://colab.research.google.com/drive/1TX-PuYCRNyoPB-WwEDqnYExtmiXsZFDx?usp=sharing

Link Dataset :

https://www.kaggle.com/code/emineyetm/telco-customer-churn/notebook

Thank you